

Question description

Alice likes festivals a lot as we all do. He also likes spending money on these festivals. He spends money in buying various things on these festivals. But he has problem of forgetting. He only remembers his top three maximum spendings for any festival.

For eg, on Holi he spends 25 units on colors, 50 units on water sprays, 100 units on gifts, 150 units on sweets but he remebers only his top 3 spendings ,i.e., 50, 100 & 150.

Now as the year ends he wants to know the **festival on which he spent most of his money that he can remember**.

Constraints

$$1 \leq T \leq 100$$

$$1 \leq N \leq 10^4$$

$$1 \leq X \leq 10^9$$

$$1 \leq |S| \leq 10; |S| \text{ denotes string length}$$

S will consist of **lowercase and/or uppercase English aplhabets**

Input Format

First line contains an integer **T**, denoting number of test cases.

Second line contains an integer **N**, denoting number of spendings in a year.

Each of the next **N** lines contain a string **S** and an integer **X** denoting festival name and spending on one of the events of that festival.

Output Format

For each test case, print a single line containing the festival name and its total spending that he remembers.

If there are more than 1 festival with the maximum spending, print the one which has lexicographically smallest name.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

2
6
B 20
A 2
A 10
A 10
B 30
A 30
3
abc 10
xyz 15
oop 8

EXPECTED OUTPUT

A 50
xyz 15

Test Case 2

INPUT (STDIN)

2
5
BA 25
AD 24
Ag 10
AE 15
AT 30
3
abc 15
xyz 25
oops 8

EXPECTED OUTPUT

AT 30
xyz 25

▼ Mandatory Test Cases

Test Case 1

KEYWORD

while(t--)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

260

Test Case 3

NLOC

46

Hashing3.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 #define MAX_FEST 10005
6 #define NAME_LEN 15
7
8 typedef struct {
9     char name[NAME_LEN];
10    long long top3[3];
11    int count;
12} Festival;
13
14int cmp_desc(const void *a, const void *b) {
15    long long v1 = *(const long long *)a;
16    long long v2 = *(const long long *)b;
17    if (v2 > v1) return 1;
18    if (v2 < v1) return -1;
19    return 0;
20}
21
22int find_or_add(Festival *festivals, int *num_festivals, const char *name) {
23    for (int i = 0; i < *num_festivals; i++) {
24        if (strcmp(festivals[i].name, name) == 0) {
25            return i;
```

```

26     }
27 }
28 strcpy(festivals[*num_festivals].name, name);
29 festivals[*num_festivals].count = 0;
30 festivals[*num_festivals].top3[0] = 0;
31 festivals[*num_festivals].top3[1] = 0;
32 festivals[*num_festivals].top3[2] = 0;
33 (*num_festivals)++;
34 return (*num_festivals) - 1;
35}
36
37void add_spending(Festival *f, long long amount) {
38     if (f->count < 3) {
39         f->top3[f->count++] = amount;
40         qsort(f->top3, f->count, sizeof(long long), cmp_desc);
41     } else {
42         if (amount > f->top3[2]) {

```

```
43         f->top3[2] = amount;
44         qsort(f->top3, 3, sizeof(long long), cmp_desc);
45     }
46 }
```

main | CH.SC.U4CSE25225 | Hashing3.c

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```
48
49int main() {
50    int t;
51    if (scanf("%d", &t) != 1) return 0;
52
53    Festival *festivals = (Festival *)malloc(MAX_FEST * sizeof(Festival));
54
55    while(t--) {
56        int n;
57        if (scanf("%d", &n) != 1) break;
58
59        int num_festivals = 0;
```

```

60     for (int i = 0; i < n; i++) {
61         char name[NAME_LEN];
62         long long amount;
63         scanf("%s %lld", name, &amount);
64         int idx = find_or_add(festivals, &num_festivals, name);
65         add_spending(&festivals[idx], amount);
66     }
67
68     char best_name[NAME_LEN] = "";
69     long long best_sum = -1;
70
71     for (int i = 0; i < num_festivals; i++) {
72         long long sum = 0;
73         for (int j = 0; j < festivals[i].count; j++) {
74             sum += festivals[i].top3[j];
75         }
76         if (sum > best_sum) {

```

```
77         best_sum = sum;
78         strcpy(best_name, festivals[i].name);
79     } else if (sum == best_sum) {
80         if (strcmp(festivals[i].name, best_name) < 0) {
81             strcpy(best_name, festivals[i].name);
82         }
83     }
84 }
85
86 printf("%s %lld\n", best_name, best_sum);
87 }
88
89 free(festivals);
90 return 0;
91}
92
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2
```

```
6
```

```
B 20
```

```
A 2
```

```
A 10
```

```
A 10
```

```
B 30
```

```
A 30
```

```
3
```

```
abc 10
```

```
xyz 15
```

```
oop 8
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing3.exe
```

```
A 50
```

```
xyz 15
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
5
```

```
BA 25
```

```
AD 24
```

```
Ag 10
```

```
AE 15
```

```
AT 30
```

```
3
```

```
abc 15
```

```
xyz 25
```

```
oops 8
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing3.exe
```

```
AT 30
```

xyz 25

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>